Experiment 2 - Conditionals and Iterative Loops

Python Lab Manual — Program and Output

EX. NO: 2(a) — Number series

PROGRAM

```
n = int(input("Enter the number of terms for the series: "))
i = 1
x = 0

while i <= n:
    x = x * 2 + 1
    print(x)
    i += 1
```

OUTPUT

```
Enter the number of terms for the series: 5
1
3
7
15
31
```

EX. NO: 2(b) — Number patterns

PROGRAM

```
rows = int(input("Enter the number of rows: "))

for i in range(1, rows + 1):
    for j in range(i):
        print(i, end=" ")
    print()
```

OUTPUT

```
Enter the number of rows: 5
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

EX. NO: 2(c) — Pyramid

PROGRAM

```
n = int(input("Enter the number of rows: "))

for i in range(n):
    for j in range(i + 1):
        print("*", end="")
    print()
```

OUTPUT

```
Enter the number of rows: 5
*
**
***
****
*****
```